

Question Label : Multiple Choice Question

Which of the following could the course team of Design Thinking for Data-Driven App Development do to make the upcoming term even more legendary?

(Hint: Our team wants to give you 1 mark as a token of thanks for being awesome. All statements are correct.) [Check 'All' as answer]

1. Send each team free pizza during their late-night GP1, GP2, GP3 marathons.
2. Invite Elon Musk (or at least his Twitter bot) to the final presentations.
3. Add a "Best Meme Prototype" award alongside "Best App Idea."
4. Give participants a lifetime supply of sticky notes so they never run out during brainstorming.

Select the correct answer from the options below:

Options :

6406534766247. ✖ Only 1

6406534766248. ✖ Only 2

6406534766249. ✖ Only 3

6406534766250. ✔ All.

Game Theory and Strategy

Section Id :	64065399926
Section Number :	15
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653230875
Question Shuffling Allowed :	No

Question Number : 335 Question Id : 6406531426157 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534766251. ✓ YES

6406534766252. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653230876

Question Shuffling Allowed :

No

Question Id : 6406531426158 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (336 to 337)

Question Label : Comprehension

Three machines, A, B, and C, produce respectively 50%, 30% and 20% of the total no. of items of a factory. The percentages of defective output of these machines are 3%, 4% and 5%.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 336 Question Id : 6406531426159 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

If an item is selected at random, the probability that the item is defective is _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.035 to 0.04

Question Number : 337 Question Id : 6406531426160 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Suppose an item is selected at random and found to be defective. The probability that the item was produced by machine A will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.40 to 0.405

Question Id : 6406531426168 **Question Type :** COMPREHENSION **Sub Question Shuffling**

Allowed : No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (338 to 339)

Question Label : Comprehension

Two roommates need to each choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j , then the player i 's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 338 **Question Id :** 6406531426169 **Question Type :** SA

Correct Marks : 2

Question Label : Short Answer Question

What is the best response of roommate 2 if roommate 1 chooses $t_1 = 0$?

$t_2 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 339 **Question Id :** 6406531426170 **Question Type :** SA

Correct Marks : 2

Question Label : Short Answer Question

If the unique, symmetric Nash equilibrium in pure strategy for this game

is represented by $t_1 = t_2 = \frac{\alpha}{3}$

then the value of α will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Id : 6406531426171 **Question Type :** COMPREHENSION **Sub Question Shuffling**

Allowed : No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (340 to 347)

Question Label : Comprehension

A market game has two buyers and one seller. Seller 1 has a watch whose worth for him is ₹200 and he offers it for sale. Players 2 and 3 are two potential buyers who do not own a watch. The worth of watch is ₹280 for one buyer (player 2) and ₹300 for the other buyer (player 3). Each buyer is interested in paying the lowest price for the watch and unwilling to pay more than the watch's worth to him. Describe the game in coalition form and answer the given sub questions.

Sub questions

Question Number : 340 **Question Id :** 6406531426172 **Question Type :** SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(1) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

200

Question Number : 341 **Question Id :** 6406531426173 **Question Type :** SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(2) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 342 Question Id : 6406531426174 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(3) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 343 Question Id : 6406531426175 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(\phi) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 344 Question Id : 6406531426176 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(2, 3) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 345 Question Id : 6406531426177 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$$v(1, 2) = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

280

Question Number : 346 Question Id : 6406531426178 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$$v(1, 3) = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

300

Question Number : 347 Question Id : 6406531426179 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$$v(1, 2, 3) = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

300

Sub-Section Number :

3

Sub-Section Id :

640653230877

Question Shuffling Allowed :

No

Question Id : 6406531426161 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (348 to 353)

Question Label : Comprehension

Consider the market for a single network good as discussed in the lecture, and write a different consumer's utility function for joining the network as $U(\theta)=\theta(a+vn^e)$ capturing the idea that it may be more plausible that a user who has a higher value for the standalone component of a technology also assigns more importance to the size of its network (i.e., consumers differ in their valuation of both the standalone and the network benefits). Here, a is the standalone benefit, $v > 0$ measures the network effect, n^e is the expected number of users joining the network and θ is uniformly distributed on the unit interval.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 348 Question Id : 6406531426162 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The value $\hat{\theta}$ corresponding to an indifferent consumer for a given price p and a given expected network size n^e will be

Options :

6406534766255. ✖ $\frac{p-a}{vn^e}$

6406534766256. ✔ $\frac{p}{(a+vn^e)}$

6406534766257. ✖ $\frac{ap}{(a+vn^e)}$

6406534766258. ✖ None

Question Number : 349 Question Id : 6406531426163 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The maximum value of p to have buyers will be

Options :

6406534766259. ✔ $a + vn^e$

6406534766260. ✖ vn^e

6406534766261. ✖

$$\frac{a}{vn^e}$$

6406534766262. ✖ $\frac{a}{1+vn^e}$

Question Number : 350 Question Id : 6406531426164 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The mass of buyers (n) as a function of $\hat{\theta}$ will be

Options :

6406534766263. ✖ $\hat{\theta}$

6406534766264. ✖ $1 + \hat{\theta}$

6406534766265. ✔ $1 - \hat{\theta}$

6406534766266. ✖ $\frac{1+\hat{\theta}}{1-\hat{\theta}}$

Question Number : 351 Question Id : 6406531426165 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The willingness to pay ($p(n^e, n)$) for the n^{th} unit of the good when n^e units are expected to be sold will be

Options :

6406534766267. ✖ $(a + vn^e)$

6406534766268. ✖ $n(a + vn^e)$

6406534766269. ✖ $(a + vn^e)(1 + n)$

6406534766270. ✔ $(a + vn^e)(1 - n)$

Question Number : 352 Question Id : 6406531426166 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

$p(n^e, n)$ is _____ (more than one options are correct)

Options :

6406534766271. ✖ increasing function of n

6406534766272. ✔ decreasing function of n

6406534766273. ✖ decreasing function of n^e

6406534766274. ✔ increasing function of n^e

Question Number : 353 Question Id : 6406531426167 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The fulfilled-expectations demand curve in this setting will be

Options :

6406534766275. ✖ $(a + vn)$

6406534766276. ✖ $n(a + vn)$

6406534766277. ✖ $(a + vn)(1 + n)$

6406534766278. ✔ $(a + vn)(1 - n)$

Question Id : 6406531426180 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (354 to 357)

Question Label : Comprehension

Calculate the Shapley value of the following game using the procedure for dissolving a partnership

$$N = \{1, 2, 3, 4\}$$

$$v(1) = 24$$

$$v(2) = 48$$

$$v(3) = 24$$

$$v(4) = 72$$

$$v(1, 2) = 96$$

$$v(2, 3) = 144$$

$$v(1, 4) = 120$$

$$v(1, 3) = 0$$

$$v(2, 4) = 0$$

$$v(3, 4) = 0$$

$$v(1, 3, 4) = 168$$

$$v(1, 2, 3) = 0$$

$$v(1, 2, 4) = 0$$

$$v(2, 3, 4) = 0$$

$$v(1, 2, 3, 4) = 240$$

$$v(\emptyset) = 0$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 354 Question Id : 6406531426181 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

$S_1 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

74

Question Number : 355 Question Id : 6406531426182 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

$S_2 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 356 Question Id : 6406531426183 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

$S_3 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

62

Question Number : 357 Question Id : 6406531426184 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

$S_4 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

74

Sub-Section Number : 4

Sub-Section Id : 640653230878

Question Shuffling Allowed : No

Question Id : 6406531426192 **Question Type :** COMPREHENSION **Sub Question Shuffling Allowed :** No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (358 to 360)

Question Label : Comprehension

Three firms are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$ where n is the number of firms that enter. The cost of entering is 62.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 358 **Question Id :** 6406531426193 **Question Type :** SA

Correct Marks : 2

Question Label : Short Answer Question

How many pure strategy equilibria are there in this game?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 359 **Question Id :** 6406531426194 **Question Type :** MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In one of these pure NEs, all three players enter the market

Options :

6406534766305. ✖ TRUE

6406534766306. ✔ FALSE

Question Number : 360 Question Id : 6406531426195 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

There is a symmetric mixed strategy equilibrium where all three players enter with the same probability (say p). Find the value of p

Options :

6406534766307. ✖ 1/5

6406534766308. ✖ 3/5

6406534766309. ✔ 4/5

6406534766310. ✖ 2/5

Sub-Section Number :

5

Sub-Section Id :

640653230879

Question Shuffling Allowed :

No

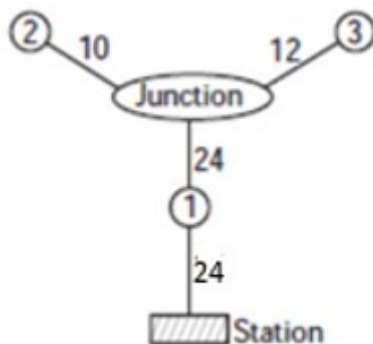
Question Id : 6406531426185 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (361 to 363)

Question Label : Comprehension

A cable TV station wants to connect three new customers to its network. The connections to the network are described in the following tree graph:



Calculate the distribution of the costs among the three players if all three decide to connect to the network.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 361 Question Id : 6406531426186 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 1 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8

Question Number : 362 Question Id : 6406531426187 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 2 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 363 Question Id : 6406531426188 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 3 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

32

Question Id : 6406531426189 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (364 to 365)

Question Label : Comprehension

Five bidders with private valuations uniformly distributed between 0 and 1 participate in a sealed bid first price auction. Assuming that everyone is playing the symmetric equilibrium bidding strategy answer the given sub-questions:

Sub questions

Question Number : 364 Question Id : 6406531426190 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Optimal bid for a bidder with valuation 0.75 will be

Options :

6406534766296. ✖ 0.7

6406534766297. ✔ 0.6

6406534766298. ✖ 0.5

6406534766299. ✖ 0.75

Question Number : 365 Question Id : 6406531426191 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

The expected revenue of the seller in this auction scenario will be

Options :

6406534766300. ✖ 4/5

6406534766301. ✖ 3/5

6406534766302. ✔ 2/3

6406534766303. ✖ 2/5

Question Id : 6406531426199 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (366 to 368)

Question Label : Comprehension

Suppose a search engine has two ad slots that it can sell. Slot **a** has a clickthrough rate of 10 and slot **b** has a clickthrough rate of 5. Three advertisers are interested in these slots. Advertiser **x** values click at 3 per click, advertiser **y** values click at 2 per click, and advertiser **z** values click at 1 per click.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 366 Question Id : 6406531426200 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Total valuation of socially optimal allocation will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

40

Question Number : 367 Question Id : 6406531426201 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

VCG price for slot **a** will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

15

Question Number : 368 Question Id : 6406531426202 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

VCG price for slot **b** will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number :

6

Sub-Section Id :

640653230880

Question Shuffling Allowed :

No

Question Id : 6406531426196 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (369 to 370)

Question Label : Comprehension

A man dies, leaving an estate worth 500 units. The deceased has two creditors; one's claim is 400 units, and the other's claim, 300 units. The division of the estate between them is carried out according to the contested garment principle.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 369 Question Id : 6406531426197 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Creditor 1 with claim 400 units gets

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

300

Question Number : 370 Question Id : 6406531426198 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Creditor 2 with claim 300 units gets

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

200

Sub-Section Number :

7

Sub-Section Id :

640653230881

Question Shuffling Allowed :

Yes

Question Number : 371 Question Id : 6406531426203 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Suppose 5 numbers are drawn independently from the uniform distribution on the interval $[0, 1]$ and then sorted from smallest to largest. The expected value of the number in second position on the list is

Options :

6406534766316. ✖ 5/6

6406534766317. ✖ 2/3

6406534766318. ✖ 1/2

6406534766319. ✔ 1/3

Sub-Section Number :

8

Sub-Section Id :

640653230882

Question Shuffling Allowed :

Yes

Question Number : 372 Question Id : 6406531426204 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In weighted majority game [7; 1, 4, 3, 5]

Options :

6406534766320. ✓ Player 1 is null player

6406534766321. ✗ Player 2 is null player

6406534766322. ✗ Player 3 is null player

6406534766323. ✗ Player 4 is null player

Question Number : 373 Question Id : 6406531426205 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Consider following two statements:

1. Crowd can be wrong even if everyone in the crowd is rational and everyone takes the same action.
2. Aggregate behaviour of many people with limited information can sometimes produce very accurate results.

Which statement(s) is (are) true?

Options :

6406534766324. ✗ Only 1

6406534766325. ✗ Only 2

6406534766326. ✓ Both 1 & 2

6406534766327. ✗ None

Question Number : 374 Question Id : 6406531426206 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct alternative

Options :

6406534766328. ✗ A stable matching system is necessarily a system under which everyone is satisfied.

6406534766329. ✗ A matching system is stable when an unmatched pair may find it beneficial to deviate from the matching and form their own match.

6406534766330. ✗ Both A stable matching system is necessarily a system under which everyone is satisfied & A matching system is stable when an unmatched pair may find it beneficial to deviate from the matching and form their own match.

6406534766331. ✓ None